MILITARY STANDARD

RAILCAR LOADING OF HAZARDOUS MATERIALS

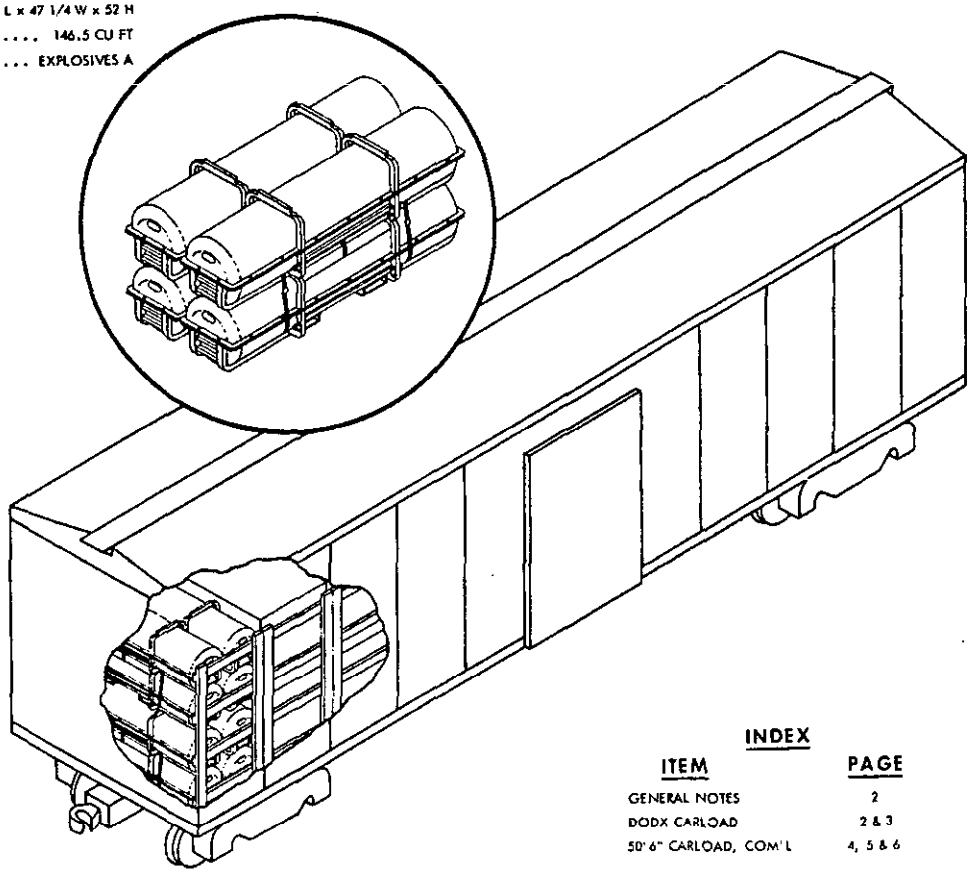
DISPENSER AND BOMB,AIRCRAFT CBU-MK 20 MODS (ROCKEYE II) IN CONTAINER MK 427 MOD 1 FLEET ISSUE UNIT LOAD

MIL-STD-1325-174
(NAVY)

24 JANUARY 1975

UNIT LOAD DATA

UNIT LOAD DRAWING WR-54/266
WEIGHT (APPROXIMATE)
DODIC/NALC NO. E173 ..... 3898 LBS
DODIC/NALC NO. E819 ..... 3926 LBS
DIMENSIONS ..... 103 L x 47 1/4 W x 52 H
CUBE ..... 146.5 CU FT
DOT HAZARD CLASSIFICATION ..... EXPLOSIVES A



INDEX

ITEM	PAGE
GENERAL NOTES	2
DODX CARLOAD	2 & 3
50'6" CARLOAD, COM'L	4, 5 & 6

NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCK-LOADING AND CONTAINERLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY).

FSC 8140

AUTHORIZED AND RELEASED FOR GENERAL USE.

M.S. Gray 1/6/75
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J.E. Kelly 1/6/75
SIGNATURE AIRSYSCOM BY DIRECTION DATE

APPROVED BY BUREAU OF EXPLOSIVES

A.F. Grasmuck 12/12/74
SIGNATURE SUPERVISOR, MILITARY & INTERMODEL SERVICES DATE

ORIGINATOR Charles McBride 12/26/74
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

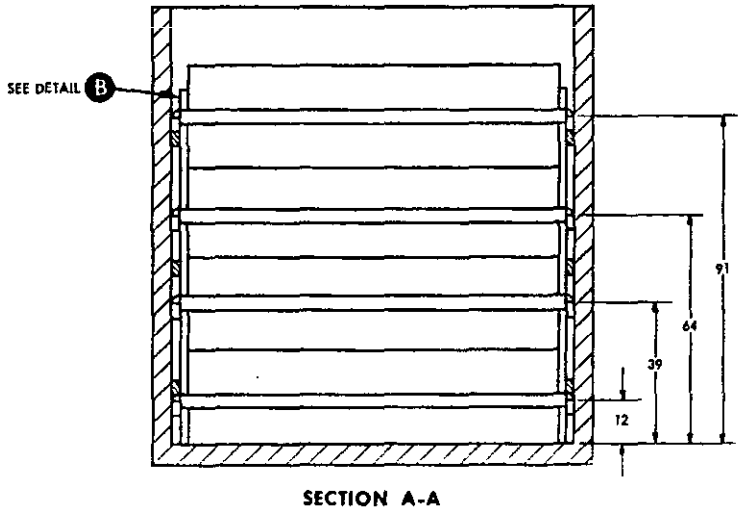
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GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS MATERIALS".
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT. UTILITY LOADER CARS SHALL BE SERIES DODX 28000.
3. LOADING PLANS SHOWN ARE FOR DODX UTILITY LOADER CAR WITH 50 FT 6 INCHES INSIDE LENGTH, 107 3/4 INCHES INSIDE WIDTH BETWEEN RAILS (111 INCHES INSIDE WIDTH BETWEEN SIDE WALLS), AND COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH INCLUDING ALL METAL CONSTRUCTED BOXCARS.
4. THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. THE LOAD CONSISTS OF AIRCRAFT DISPENSER AND BOMB, CBU-MK 20 & MODS (ROCKEYE II) IN CONTAINER MK 427 MOD 1. THE CONTAINERS ARE UNITIZED FOUR TO A UNIT LOAD AS SHOWN IN WR-54/266.
7. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY).
9. ALL STRAP JOINTS SHALL BE CRIMP-TYPE JOINTS MADE WITH A CRIMPING TOOL THAT DOES NOT CUT THE SEAL OR STRAP, RATHER THAN A TOOL THAT MAKES A NOTCH-TYPE JOINT. THE STRAPPING MUST BE IN ACCORDANCE WITH NAVSUP NOTICE 4642 DATED 21 APRIL 1971.
10. APPLICABLE MATERIAL SPECIFICATIONS:
    DUNNAGE LUMBER - FED. SPEC MM-L-751
    NAILS - FED. SPEC FF-N-105
    STRAPPING - FED. SPEC QQ-S-781, TYPE I, HEAVY DUTY, CLASS A, DRY (UNLUBRICATED).
    SEALS - FED. SPEC QQ-S-781, STYLE III, HEAVY DUTY
11. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA, CLOSE AND SEAL BOXCAR DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED IN OP 2165 (VOL. I).

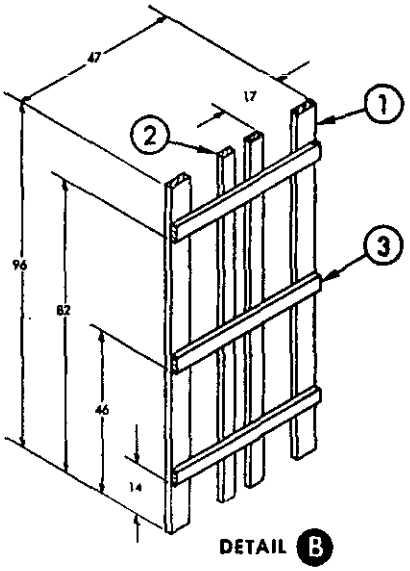
50 FT 6 IN. BOXCAR, DODX

1. THE CARLOAD CONSISTS OF 22 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. A DETAILED DESCRIPTION AND OPERATING INSTRUCTIONS FOR THE UTILITY LOADER ARE CONTAINED IN OP 1750.
3. TO PREVENT UNUSED "DF" EQUIPMENT FROM BECOMING DISLODGED DURING TRANSIT OF DODX CARS SECURE IT AT ANY LOCATION IN THE BOXCAR WHICH WILL NOT INTERFERE WITH UNLOADING.
4. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN DODX BOXCARS THE SAME PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE. ANY BAYS OR PORTION THEREOF MAY BE USED PROVIDING THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR ARE COMPLIED WITH, SEE MIL-STD-1325 (NAVY). EACH CROSSMEMBER WILL BE USED IN SUCH A MANNER THAT IT WILL RETAIN NOT MORE THAN 2000 LBS OF THE LADING.



SECTION A-A

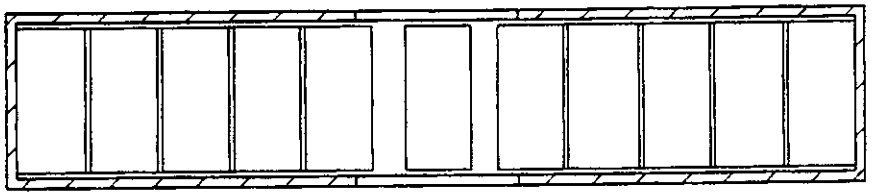
SHOWING LOCATIONS OF WALL MEMBERS USED FOR CROSSMEMBERS



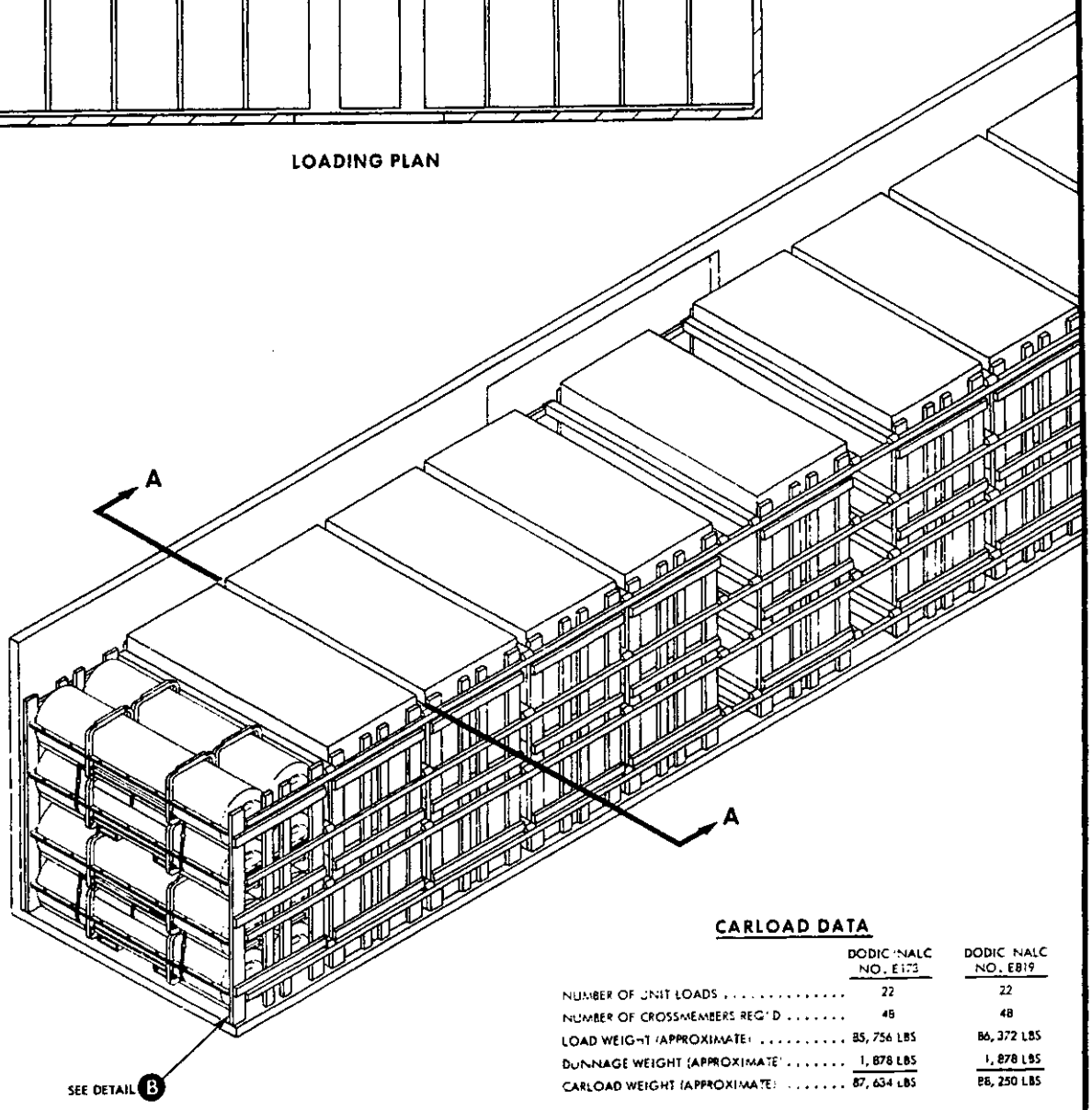
DETAIL B

SWAY BRACE
22 REQUIRED

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LOADING PLAN



CARLOAD DATA

	DODIC NALC NO. E173	DODIC NALC NO. E819
NUMBER OF UNIT LOADS	22	22
NUMBER OF CROSSMEMBERS REQ'D	48	48
LOAD WEIGHT (APPROXIMATE)	85, 756 LBS	86, 372 LBS
DUNNAGE WEIGHT (APPROXIMATE)	1, 878 LBS	1, 878 LBS
CARLOAD WEIGHT (APPROXIMATE)	87, 634 LBS	88, 250 LBS

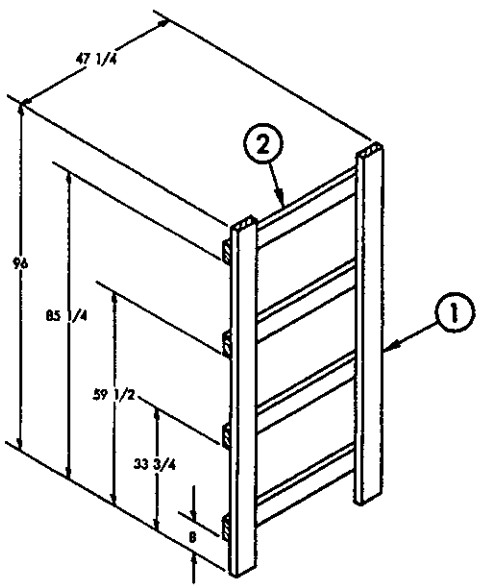
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE NAILS
3	SWAY BRACE HORIZONTAL	2 x 4 x 46	66	SEE 1, 2	-	-
2	SWAY BRACE VERTICAL	2 x 4 x 96	44	3	3 /JOINT	10d
1	SWAY BRACE VERTICAL	2 x 6 x 96	44	3	3 /JOINT	10d

LIST OF MATERIALS AND NAILING DATA

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50 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 24 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH, SEE MIL-STD-1325 (NAVY).
3. WHEN BOXCARS WIDER THAN 9 FT. 2 IN. ARE USED, INCREASE THE THICKNESS OF PIECE 1 BY LAMINATING AN APPROPRIATELY THICK PIECE TO IT. IF THE WIDTH OF BOXCARS USED IS LESS THAN 9 FT. 2 IN. DECREASE THE THICKNESS OF PIECE 1 AS NECESSARY.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG DOORS. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.



DETAIL A

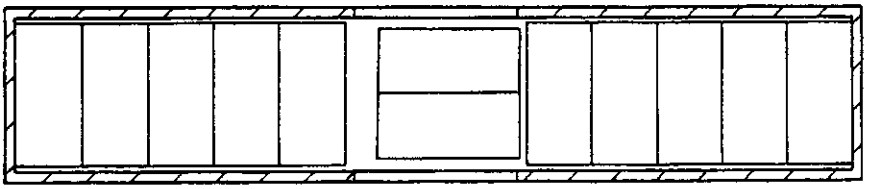
SWAY BRACE
20 REQUIRED

*2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

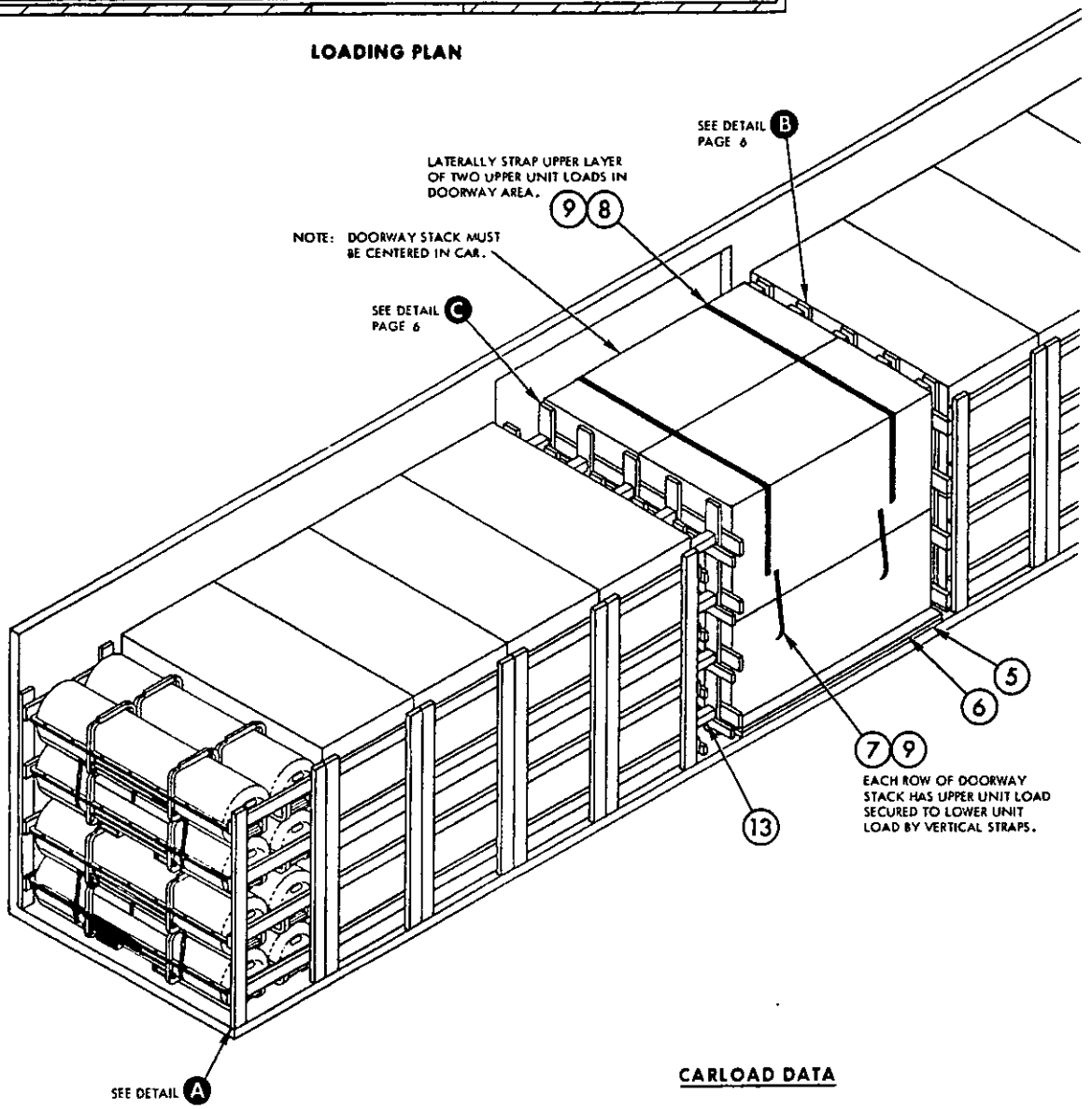
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
14	CENTER GATE HOLD DOWN	2 x 3 x 48	2	11	2 PER JOINT	10d
13	STRUT	4 x 4 x WEDGE FIT*	20	11	2 PER JOINT	16d
12	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	11	3 PER JOINT	10d
11	CENTER GATE VERTICAL	2 x 6 x 104	10	SEE 10	-	-
10	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	8	11	3 PER JOINT	10d
9	SEAL	1 1/4	12	-	-	-
8	HORIZONTAL STRAP	1 1/4 x .035 x 22 FT	2	-	-	-
7	VERTICAL STRAP	1 1/4 x .035 x 12 FT 6	4	-	-	-
6	SLEEPER	2 x 6 x 103	2	5	1 PER FT	16d
5	SLEEPER	2 x 6 x 103	2	CAR FLOOR	1 PER FT	16d
4	SEPARATOR GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	4	SEE 3	-	-
3	SEPARATOR GATE VERTICAL	2 x 6 x 104	10	4	3 PER JOINT	10d
2	SWAY BRACE HORIZONTAL	2 x 6 x 47 1/4	80	1	3 PER JOINT	10d
1	SWAY BRACE VERTICAL	2 x 6 x 96	40	SEE 2	-	-

LIST OF MATERIALS AND NAILING DATA

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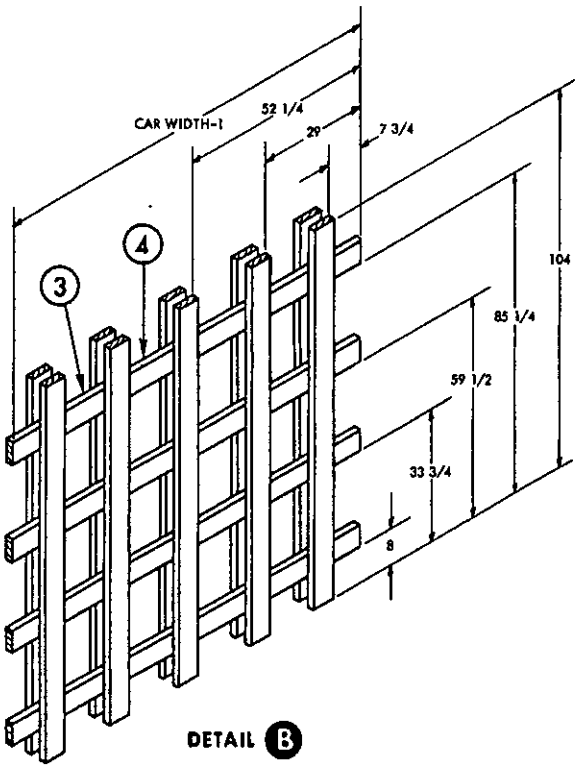
LOADING PLAN



CARLOAD DATA

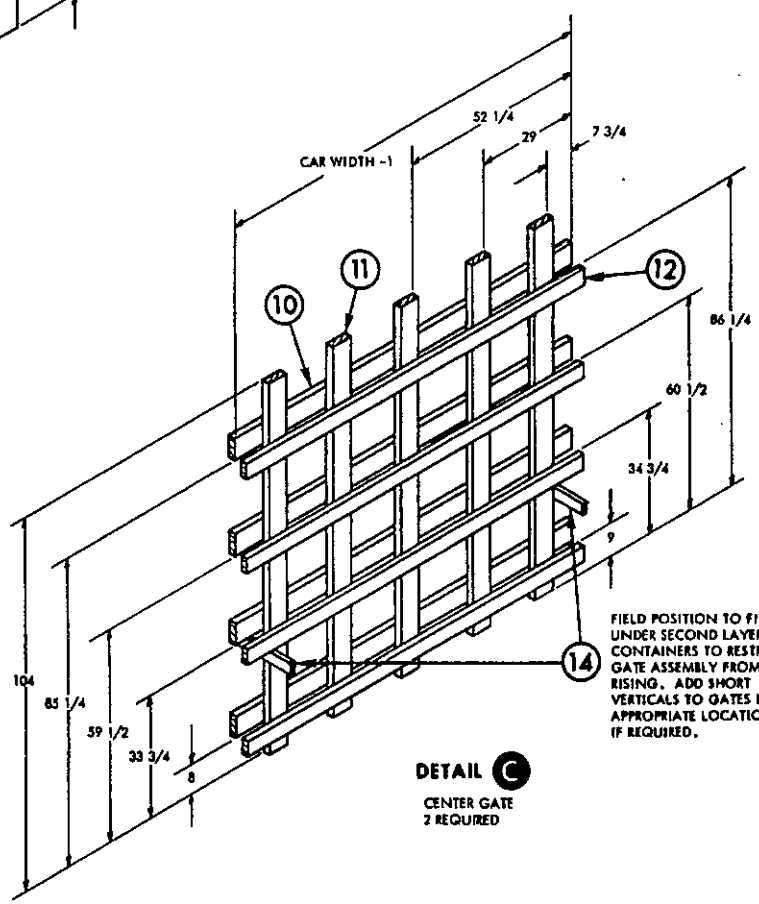
	DODIC/NALC NO. E173	DODIC/NALC NO. E819
NUMBER OF UNIT LOADS	24	24
LOAD WEIGHT (APPROXIMATE)	93,552 LBS	94,224 LBS
DUNNAGE WEIGHT (APPROXIMATE)	2,645 LBS	2,645 LBS
CARLOAD WEIGHT (APPROXIMATE)	96,197 LBS	96,869 LBS

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DETAIL B

SEPARATOR GATE
1 REQUIRED



FIELD POSITION TO FIT UNDER SECOND LAYER CONTAINERS TO RESTRAIN GATE ASSEMBLY FROM RISING. ADD SHORT VERTICALS TO GATES IN APPROPRIATE LOCATIONS IF REQUIRED.

DETAIL C

CENTER GATE
2 REQUIRED

REVIEW ACTIVITY:
NAVY - OS, AS

PREPARING ACTIVITY:
NAVY - OS
(PROJECT NO. 8140-N264)

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